Evaluating Complaint Submission and Status Comprehension in a Citizen Service Portal

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Abstract. A complaint portal helps only if citizens can send a complete report and understand what the displayed status means for them. This paper evaluates both abilities in GovSmart, a prototype portal for citizen complaints to a local government. We ask whether users can submit a complete complaint and understand the next action from the status shown. In a pilot with [N] prospective users, each participant performed three tasks (submit, revise, and find the status of a report) from a synthetic road-damage scenario and explained five report statuses against a prepared answer key. We also tested input validation and report ownership with two citizen accounts. Outcome, time, help, and errors were logged per participant and task. [X of N submissions were complete without help; Y of Z status interpretations were correct, and the most misread term was T; V of W validation and ownership cases passed.] Failures were traced to [specific form parts and status terms] and linked to interface revisions. The pilot is small and uses synthetic reports, so it supports problem identification only. It does not show higher citizen trust or better service quality of any agency.

Keywords: Citizen complaint portal · Usability evaluation · Task success · Status comprehension · E-government · Form validation.

1 Introduction

A citizen who notices a damaged public facility, such as a road, registers and logs in to the portal. The citizen writes a title and a description, may add the coordinates of the location, and may attach a photo as evidence. The portal assigns the report to a department and an urgency level with keyword-based routing and stores it with the status *diterima* ("received"). While the report has this status its owner can edit or cancel it. Afterwards, department staff update the status, may leave an officer note, and may upload a feedback document that the citizen can open. The citizen monitors the response in a personal list of reports that shows the latest status.

Forms and status displays can create barriers for citizens, but we do not assume that these barriers hold for all citizens. We ground the problem statement

in observations of the first tasks of our pilot: [describe the form fields at which participants hesitated or erred and the status terms they misread, with counts per field and per term]. Two design properties of the prototype motivated the checks reported here. The status is shown to citizens as the lowercase token returned by the API (for example *diterima*), and the evidence photo is sent in a second request after the report itself has been saved, so a report can exist without its attachment.

Complaint portals have mostly been evaluated through satisfaction surveys [1, 2], qualitative implementation studies [3], or field usability tests with a small number of citizens [4]. Research on web forms studies completion and error handling [5, 6]. In our search we found no study of a complaint portal that asks participants to explain, in their own words, what a status means and what they must do next, although keeping users informed about system status is a long-standing usability heuristic [7]. Our work also differs from SIGAP, a companion prototype study in the same course that examines the internal consistency of workflow transitions and assignment control on the server side **[TODO: cite SIGAP as agreed with the lecturer]**. We examine instead what a citizen can do and understand with the portal.

This paper makes a citizen-side usability evaluation of the GovSmart portal. Its contributions are: (i) a task-based pilot on complete submission, revision, and status finding; (ii) a status-comprehension test against a written answer key; and (iii) a check of input validation and report ownership, with each proposed interface improvement linked to an error that we observed. We answer one research question: *can users submit a complete complaint and understand the next action from the status shown?*

2 Related Work

2.1 Evaluation of Citizen Complaint Portals

The Kaohsiung Citizen Electronic Complaint System was assessed with satisfaction and service quality as indicators, and the ability to resolve complaints contributed most to overall satisfaction [1]. A user survey of the Indonesian national channel SP4N-LAPOR! with 2,420 respondents reported, among other indicators, that 90.8% rated the simplification of complaint submission positively [2]. Both measure opinion after use rather than what users can do. A qualitative study of the Lapak Aduan service in Banyumas, based on observation, interviews, and documentation, found shortcomings in usefulness and ease of use and low utilisation [3]. Closest to our design, field tests of the Pakistan Citizens' Portal app measured efficiency and effectiveness following ISO 9241-11 [8] and satisfaction with a questionnaire, with more than a dozen citizens, and located weaknesses in registration and findability [4].

2.2 Form Completion

Form guidelines ask that labels and input fields tell users what data is expected and that errors are shown clearly [5]. An eye-tracking study of a four-step web

form with 32 participants compared four locations of error messages; the researcher did not intervene during form filling, and the group differences in completion time were not significant, while participants preferred format restrictions, examples in error messages, and immediate feedback [6]. An earlier study with 303 participants found messages placed next to the erroneous field most effective [9]. These studies use generic forms, not complaint reports with evidence attachments.

2.3 Status Comprehension

Keeping users informed about what is going on is one of the usability heuristics [7], but a status is only informative if its wording is understood. Label-comprehension studies exist in other domains; for example, studies of app privacy labels identified widely misunderstood technical terms [10]. We did not find a comparable study for complaint statuses, so we test it directly (Sect. 3.4).

2.4 Comparison

Table 1 compares the studies on participants, tasks, moderator help, and success metrics, the four aspects named in our evaluation plan.

3 Methodology

3.1 System Under Evaluation

GovSmart is a web portal with a React front end and a FastAPI back end on a MySQL database. The front-end–back-end flow of submission, revision, and status display was completed and frozen before the study (version **[TODO: commit hash and date]**). A report has a title of 5–100 characters, a description of 20–2,000 characters, an optional location (latitude in $[-90, 90]$, longitude in $[-180, 180]$), and an optional photo (JPEG, PNG, or WEBP, at most 5 MB, sent in a separate request). A new report receives the status *diterima*. A citizen can edit or cancel a report only while it is *diterima*, can read only own reports, and sees the officer note, when present, on the report card. The personal report list returned by the API holds at most five reports.

3.2 Participants, Consent, and Digital Experience

We recruit [5–8] prospective users, [N] in the end, who are not members of the project team. Each participant gives informed consent after hearing the purpose, the recorded data, and the right to stop at any time. Participants are identified as P1–P[N] only, and the study uses synthetic accounts and synthetic reports, never real identity numbers. Before the tasks we record age band, main device, how often the participant uses online government or other transaction services, and any earlier experience of filing a complaint. We apply the user-study protocol of the course guide **[TODO: add the protocol items from the first pages of the guide, which were not in the file we used]**.

Table 1. Comparison of related studies. Entries follow the abstracts or summaries of the sources; "n/s" = not stated there.

Study	Participants	Tasks	Moderator help	Success metrics
Yeh and Chu [1]	Citizens (survey)	None (retrospective survey)	n/a	Satisfaction, service quality
UNDP survey [2]	2,420 respondents who used SP4N-LAPOR!	None (survey)	n/a	Share rating indicators positively
Indrayani et al. [3]	n/s	Observation and interviews	n/s	Qualitative findings on usefulness, ease of use
Dahri et al. [4]	More than a dozen citizens	Field tests of app screens, incl. lodging a complaint	n/s	Efficiency, effectiveness (ISO 9241-11), satisfaction
Inal [6]	32 students	Four-step web form	None (no intervention)	Completion, error recognition, error correction time
This study	[N] prospective users	Submit, revise, find status; explain 5 statuses	Counted and coded per task	Success with/without help, time, errors, correct interpretations

3.3 Scenario, Completeness Key, and Tasks

Every participant receives the same written scenario, without naming any button or menu: a pothole has appeared on a road at a synthetic location, and a sample photo is provided. The *completeness key* for task T1 lists what a complete report must contain: a title that names the problem, a description that states street, village or district, and the condition of the road, the sample photo attached, and the correct department **[TODO: confirm the key items and whether the coordinates are required]**.

T1 – Submit. Report the problem to the local government.

T2 – Revise. After submitting, the participant is told that one detail in the description is wrong and corrects it.

T3 – Find status. In an account that holds seeded reports in different statuses, the participant finds the report named in the scenario and states its current status.

A timer starts when the participant finishes reading the instruction and stops when the correct result is reached, the time limit is reached, or the participant

Table 2. Draft answer key for the five statuses, derived from the behaviour of the prototype. The wording is a proposal to be agreed and frozen by the authors.

Status shown	Intended meaning	Who is acting	Citizen's next action
diterima	The system has received the report and routed it to a department; no officer has started work.	System, department queue	None required; the report can still be edited or cancelled.
diproses	A department officer is handling the report.	Department officer	Wait for an update; editing and cancelling are no longer possible.
selesai	The department has marked the report as handled.	No one	Read the officer note and open the feedback document, if any.
ditolak	The department declined the report; the reason is in the officer note.	No one	Read the reason; send a corrected new report if the problem remains.
dibatalkan	The citizen cancelled the report.	No one	None.

gives up. Equivalent tasks use the same time limit (**[TODO: time limit per task]**). The moderator records every help given, using a fixed list of help types, and every error. After each task we check the database for the data that were actually stored (title, description, location, photo, department, status). Outcomes are coded as success without help (S), success with help (SH), or failure (F), where F includes a wrong or incomplete stored report.

3.4 Status Comprehension Test

We choose the five statuses that exist in the portal and write the meaning and the next action for each as the answer key (Table 2). The key is agreed and frozen before the first session. Each status is shown on a concrete report card, including the officer note where the status has one. The participant explains in own words what the status means and whether any action is needed; no answer options are offered, because options would leak the answer. The participant then rates confidence from 1 to 5. Two team members code each answer as correct, partly correct, or incorrect against the key and record the terms that caused misreadings; differences are settled by discussion.

3.5 Validation and Ownership Test

With citizen accounts A and B we submit the following cases and inspect the message shown, the HTTP response, and the database: empty title and description; title and description below and above their length limits; a file of a wrong type; a file above 5 MB; an empty file; and the same submission sent twice, to

Table 3. Measures recorded.

Measure	Definition
Task outcome	S, SH, or F per participant and task (Sect. 3.3)
Time	Seconds from end of instruction to stop rule
Help and errors	Count and type per task, recorded by the moderator
Stored data	Fields actually stored versus the completeness key
Status interpretation	Correct, partly correct, or incorrect per status; misread terms
Confidence match	Confidence rating (1–5) compared with correctness
Validation and ownership	Pass or fail per case; duplicate reports; unauthorised changes; confusing messages

Table 4. Participants and conditions (output 1).

ID	Age band	Device	Online-service experience	Earlier complaint experience
P1	[]	[]	[]	[]
P2	[]	[]	[]	[]
[one row per participant; add the session date, place, and time limit per task below the table]				

check the duplicate-submission policy **[TODO: define the intended policy before the test; a report sent twice is either merged, rejected, or accepted as a second report]**. From account A we then open, edit, cancel, and attach a photo to the report of account B through the interface and through the API. Expected result: invalid input is rejected with a clear message, resubmission follows the policy, and the report of B is not accessible to A.

3.6 Data Recorded and Analysis

Table 3 lists the measures. With fewer than ten participants we report counts, medians, and the individual cases, and no significance tests. A sample of this size is suited to finding usability problems rather than to estimating rates [11]. If we build an improved version, participants perform equivalent but not identical tasks, and we discuss the learning effect; an old-versus-new comparison is not treated as a strong causal experiment.

4 Results and Discussion

4.1 Data, Cases, and Conditions

Table 4 describes the participants and the experimental conditions.

Table 5. Results per category (output 2). For tasks: S = success without help, SH = success with help, F = failure. For statuses: correct, partly correct, incorrect. For validation and ownership cases: pass or fail.

Category	n	S/corr.	SH/part.	F/inc.	Med. time (s)
T1 Submit	[]	[]	[]	[]	[]
T2 Revise	[]	[]	[]	[]	[]
T3 Find status	[]	[]	[]	[]	[]
diterima	[]	[]	[]	[]	–
diproses	[]	[]	[]	[]	–
selesai	[]	[]	[]	[]	–
ditolak	[]	[]	[]	[]	–
dibatalkan	[]	[]	[]	[]	–
Invalid input cases	[]	[]	–	[]	–
Resubmission	[]	[]	–	[]	–
Cross-account access	[]	[]	–	[]	–

Table 6. Failure examples and causes (output 3).

#	Observation (anonymised)	Form part or status term	Cause	Revision
1	[]	[]	[]	[]
2	[]	[]	[]	[]

4.2 Results per Category

Table 5 reports the results per task, per status, and per validation category.

4.3 Failure Cases and Causes

Table 6 lists the failures, each with its cause and the revision it led to.

4.4 Discussion

[Answer the research question first: how many participants submitted a complete report and found the right report without help, and how many understood the next action for each status.] [Explain each failure through the form part or status term in Table 6, for example a label, a field limit, or a lowercase status token, and say why participants read it as they did.] The validation and ownership results are reported separately from the usability results, because they concern the system rather than the participants: [state which cases failed and what the database showed]. If an improved version was tested, participants performed equivalent tasks, and part of any improvement may come from learning; we therefore treat the comparison as indicative only.

5 Conclusion

[Answer the research question with the real numbers: X of N participants submitted a complete report without help; Y of Z status interpretations were correct, with T the most misread term; V of W validation and ownership cases passed.] The pilot is small, uses synthetic reports and a convenience sample, and was run on one prototype version. It therefore supports the identification of problems in the form and in the status wording and the revisions linked to them. It does not support conclusions about citizen trust or about the service quality of any agency, because neither was measured.

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